StudyNest

Requirements Specification

A specification-aligned learning tool for students taking the T-Level in Digital Software Development.
Functional and non-functional requirements, success criteria, data model, and architecture.

Author: Mario

Document version: 2.1

Status: Working draft, intended to evolve as the build progresses

Contents

1. Document purpose and scope

This document is the requirements specification for StudyNest, a web-based learning tool that supports students preparing for the T-Level qualification in Digital Software Development. The document describes what the system must do, how it must behave, what data it manages, and what success looks like.

1.1 Purpose

The document is intended to be the single source of truth that:

- Enables a developer (or agentic AI working under supervision) to begin building the system without further clarification on what must be built.
- Provides a stable reference point against which design and implementation decisions can be checked.
- Documents acceptance criteria so that the system's first release can be objectively evaluated.
- Surfaces architectural and policy decisions early, so they are made deliberately rather than by accident during implementation.

1.2 Scope of this version

This specification describes the first usable release of StudyNest. The first release covers Core Paper 1, Core Paper 2, the Exam Technique section (with Command Verbs as its first feature), the ExplainIT teach-the-AI feature, the Revision section, and the admin interface. The Employer Set Project (ESP) and Occupational Specialism (OS) modules are described at a high level only, as later iterations.

1.3 Audience for this document

- The author and primary builder of the system.
- Any developer or contributor — human or AI-assisted — working on the codebase.
- Future collaborators evaluating whether to adopt or extend the system.
- College leadership and DPO contacts who need to understand the system's data handling and security posture.

2. Glossary

Terms used throughout this document. Where a term has both a general meaning and a project-specific meaning, the project-specific meaning is given.

Term	Definition
AO	Assessment Objective. The cognitive demand level of an exam question, ranging from AO1 (recall) through AO2 (apply) to AO3 (analyse and evaluate).
Cohort	A group of students taught together, usually corresponding to a year group at a single college (e.g. "Strode DSD Year 2 2026/27").
Command verb	A word in an exam question that signals the kind of response expected (describe, explain, analyse, evaluate, justify, discuss, compare, assess).
Core Paper	One of the two written examinations in the T-Level DSD assessment. Core Paper 1 and Core Paper 2 cover the technical content of the qualification.
DSD	Digital Software Development. The T-Level pathway StudyNest is designed for. Formerly known as Digital Production, Design and Development (DPDD).
ESP	Employer Set Project. A project-based assessment component of the T-Level. Out of scope for the first release.
Exam Technique	A top-level section of StudyNest covering exam-skill meta-content that applies across Core Paper, ESP and OS. Command Verbs is the first feature in this section.
ExplainIT	A top-level feature of StudyNest in which the student teaches a topic to an AI character that plays a confused fellow learner with plausible misconceptions, probing the student until the student demonstrates genuine understanding.
FE	Further Education. The post-16, pre-university tier of UK education in which T-Levels are delivered.
Mastery	A numerical estimate of how well a student has internalised a topic or command verb. Used to drive the spaced-repetition queue and the dashboard's recommended next session.
Misconception	A specific incorrect understanding that DSD students typically hold about a topic. Curated from real classroom observation, used to power ExplainIT.
MVP	Minimum Viable Product. The first release of StudyNest, scoped to be genuinely usable by a real cohort of students.
Pearson	The awarding body that owns the T-Level DSD specification, mark schemes, and past papers.
OS	Occupational Specialism. A project-based, vocationally-focused assessment component of the T-Level. Out of scope for the first release.
PO	Performance Outcome. A specific learning objective listed in the DSD specification that an exam question can be mapped to.
Pseudonymous account	A student account where the identifier shown in the system is a chosen handle rather than the student's real name. Used to protect privacy in cohort-level views.

Term	Definition
Spaced repetition	An evidence-based learning technique in which content is re-surfaced to a learner at intervals calibrated to how well they remembered it last time.
StudyNest	The product name. Stylised as one word with internal capital N.

3. Stakeholders and user roles

3.1 User roles

StudyNest recognises two user roles. Each role has a different set of capabilities and a different post-sign-in destination.

Role	Description and capabilities
Student	The primary end user. A T-Level DSD learner aged 16–19. After signing in: redirected to the Student Dashboard. Can: work through Core Paper topics, take quizzes, attempt extended responses, use Exam Technique and Command Verbs, run revision activities (mixed quizzes, past-question practice, spaced-repetition queue), use ExplainIT to teach the AI, view their own dashboard and progress. Cannot: see other students' data, change their own role, access admin views, create or delete accounts.
Admin	A staff user with system-management responsibilities. Initially Mario; later potentially other DSD lecturers. After signing in: redirected to the Admin Dashboard. Can: create student accounts (individually or in bulk), assign students to cohorts, view aggregate analytics across the cohort, view individual student detail, reset student passwords, deactivate student accounts, delete student accounts (with confirmation), manage system settings. Cannot (in the first release): study mode is not available to admin accounts. An admin who wants to use the study features would have a separate student account. Note: this two-role model is deliberately simple. A 'teacher' role between student and admin can be added later if other lecturers adopt StudyNest and need limited administrative access.

3.2 Stakeholders beyond direct users

- Strode College senior leadership and Data Protection Officer — StudyNest processes data about students at the college and must satisfy institutional policy.
- Pearson — the awarding body whose specification StudyNest is built around. Their position on use of mark schemes and past questions affects what is permissible.
- Other FE colleges delivering T-Level DSD — potential future adopters.
- Apprenticeship providers and employers — students may showcase StudyNest in interviews; quality reflects on the author.

4. System overview

4.1 What StudyNest is

StudyNest is a web application that helps students prepare for their T-Level Digital Software Development assessments. It is built specifically around the DSD specification, mapping every interaction — content, quizzes, feedback, and progress tracking — to the specific performance outcomes and assessment objectives that students will encounter in their exams.

StudyNest is distinguished from generic AI tutors and revision platforms by being specification-aware. Where ChatGPT will give plausible-sounding answers, StudyNest gives feedback calibrated to what DSD examiners actually reward.

4.2 What it does, in one paragraph

A student signs in to a personal dashboard that recommends what they should study next, based on what they're weakest on. They can choose to learn a Core Paper topic, take a quiz on a topic they've already studied, practise their command-verb writing in the Exam Technique section, revise across topics in the Revision section, or use ExplainIT to teach a topic to an AI character that plays a confused fellow learner. The admin (Mario) signs in to a separate Admin Dashboard with cohort analytics, student management, and system stats.

4.3 Three converging goals

Goal 1 — Student outcomes. Measurable improvement in Core Paper 1 and Core Paper 2 grade outcomes for students who use StudyNest consistently, compared to historical cohort baselines.

Goal 2 — Software engineering portfolio value. The codebase, the development workflow, and the deployed product should look and feel like real industry work, providing a portfolio asset for any students who contribute to development.

Goal 3 — Practitioner profile and recognition. Built well and used widely, StudyNest should become a distinctive contribution to the FE Digital teaching community — recognised as a named tool with a clear pedagogical philosophy.

5. Information architecture

5.1 Three surfaces

StudyNest has three distinct user-facing surfaces, each with its own purpose and audience:

Public Landing page — what an unauthenticated visitor sees. A marketing-style introduction explaining what StudyNest is, who it's for, and what it does. Includes a sign-in entry point but no protected content.

Student Dashboard — the personalised home for an authenticated student. Recommended next session, mastery overview, weak spots, recent activity. Navigation to all the learning sections.

Admin Dashboard — the home for an authenticated admin. Cohort analytics, student management, individual student detail, and system settings. Distinct from the student view; an admin does not access the study features from this dashboard.

5.2 Top-level structure

```
Public:
└── Landing page

Authenticated as student:
└── Dashboard (student)
    ├── Core Paper
    │   ├── Core Paper 1
    │   │   └── [Topic]
    │   │       ├── Concept content
    │   │       └── Topic quiz
    │   └── Core Paper 2
    │       └── [Topic]
    │           ├── Concept content
    │           └── Topic quiz
    ├── ESP                         [later iteration; nav present]
    ├── OS                          [later iteration; nav present]
    ├── Revision
    │   ├── Mixed Quizzes
    │   ├── Past-Question-Style Practice
    │   └── Spaced-Repetition Queue
    ├── ExplainIT
    │   ├── Choose a topic to teach
    │   ├── Active session
    │   └── Past sessions / transcripts
    └── Exam Technique
        └── Command Verbs
            ├── Diagnostic
            ├── Verb-by-verb Guide
            ├── Practice
            └── Quick Reference

Authenticated as admin:
└── Admin Dashboard
    ├── Cohort overview
    ├── Student management
    │   ├── Create student (individual)
    │   ├── Bulk import (CSV)
    │   ├── Reset password
    │   ├── Deactivate student
```

```
│   └── Delete student
├── Individual student detail
├── Aggregate analytics
└── Settings
```

5.3 Why this structure

- **Landing and Dashboard are separated.** The Landing page exists to introduce StudyNest to visitors and route them to sign-in. The Dashboard exists to drive personalised study for authenticated students. These are different jobs and conflating them weakens both.
- **Two distinct dashboards (student and admin) rather than one with conditional rendering.** Cleaner separation of concerns. Different audience, different content, different routes. Auth flow is simpler — the post-sign-in redirect goes to the dashboard appropriate to the role.
- **ExplainIT is top-level, not under Revision.** ExplainIT is the most distinctive feature of StudyNest. Burying it under Revision would obscure the tool's pedagogical signature. Promoting it to top-level signals that StudyNest is not just another quiz app.
- **Exam Technique is top-level.** Command-verb fluency and other exam-skill meta-content apply across Core, ESP and OS. Treating Exam Technique as a peer of the content sections signals that exam technique is a discipline of equal weight to content knowledge.
- **Exam Technique starts with Command Verbs only.** The section is designed to grow (future siblings: time management, structuring extended responses, reading the question, self-marking). The first release ships Command Verbs only; the others are noted but not yet specified.
- **Revision is separate from learning.** Topic quizzes inside Core Paper are formative knowledge checks. Revision is the cross-cutting layer for stress-testing knowledge across topics — different framing, different cognitive demand.
- **ESP and OS visible in nav now.** Even though their content is a later iteration, their nav presence sets correct user expectations and ensures the layout accommodates them from day one.

6. Functional requirements

Functional requirements describe what the system must do. Each requirement has an identifier (FR-XX) and a priority. Priority levels are:

- MUST — required for the first release. Without this, the release does not ship.
- SHOULD — strongly desired in the first release; defer only if necessary.
- COULD — included if time allows; valuable but not essential.
- LATER — out of scope for the first release; documented here so it is not forgotten.

6.1 Public Landing page

ID	Priority	Requirement
FR-01	MUST	The Landing page must be accessible without authentication and must clearly explain what StudyNest is and who it is for.
FR-02	MUST	The Landing page must include a primary call-to-action that takes the user to the sign-in page.
FR-03	MUST	The Landing page must function on mobile, tablet, and desktop viewports.
FR-04	SHOULD	The Landing page should describe the main features (Core Paper practice, Exam Technique, ExplainIT, Smart Revision) at a level a prospective student or visiting lecturer can understand.
FR-05	SHOULD	The Landing page should include an 'About this tool' section explaining StudyNest is built by an FE lecturer for FE students.

6.2 Authentication and account management

ID	Priority	Requirement
FR-10	MUST	The system must allow a user to sign in with a username (or pseudonymous handle) and password.
FR-11	MUST	The system must redirect a signed-in user to the Student Dashboard or Admin Dashboard based on their role.
FR-12	MUST	The system must require a student to change their initial password on first sign-in.
FR-13	MUST	The system must allow a user to sign out, with the session cleanly terminated.
FR-14	SHOULD	The system should rate-limit sign-in attempts to prevent brute-force attacks (5 attempts per minute per IP).
FR-15	COULD	The system could allow a student to reset their own password via an email address linked at account creation.
FR-16	LATER	The system could later support single sign-on with the college's identity provider.

6.3 Student Dashboard

ID	Priority	Requirement
FR-20	MUST	The Student Dashboard must display a personalised greeting using the student's handle.
FR-21	MUST	The Student Dashboard must show a single 'Recommended next session' card based on the student's data (weakest topic, items due in spaced-repetition queue, or recent decline).
FR-22	MUST	The Student Dashboard must show a 'Mastery overview' panel displaying mastery level for each major topic area.
FR-23	MUST	The Student Dashboard must show a 'Weak spots' panel listing 3–5 flagged items needing attention.
FR-24	SHOULD	The Student Dashboard should show a 'Recent activity' panel with the last 4–5 sessions.
FR-25	SHOULD	The Student Dashboard should show a soft consistency indicator (e.g. 'You've revised on 4 of the last 7 days').
FR-26	MUST	Each dashboard panel must link directly to the relevant action (e.g. clicking a weak topic starts a session on that topic).

6.4 Core Paper module

ID	Priority	Requirement
FR-30	MUST	The system must present each topic as a structured page containing: title, overview, specification mapping (POs and AOs), concept content, topic quiz, and (where authored) extended-response practice and misconception flags.
FR-31	MUST	The topic quiz must support multiple-choice, short-answer, and extended-response question types.
FR-32	MUST	Multiple-choice questions must be auto-graded with immediate feedback on correctness and explanation.
FR-33	MUST	Short-answer questions must be graded against a rubric using LLM-assisted scoring with explicit rubric criteria.
FR-34	MUST	Extended-response questions must receive feedback that explicitly references the command verb and at least one assessment objective.
FR-35	MUST	Each quiz attempt must be persisted (response, score, feedback, timestamp).
FR-36	MUST	The student must be able to retake a topic quiz; previous attempts remain in their history but only the latest score updates current mastery.
FR-37	SHOULD	Topic pages should display the student's current mastery level for that topic.

ID	Priority	Requirement
FR-38	SHOULD	Topic pages should suggest a next action after the quiz (e.g. 'Try a harder question', 'Review the misconceptions section').

6.5 Exam Technique → Command Verbs

Exam Technique is a top-level section. The first release contains a single feature: Command Verbs. The section is structured to accommodate future siblings (time management, structuring extended responses, reading the question, self-marking), but those are out of scope here.

ID	Priority	Requirement
FR-40	MUST	The Exam Technique section must present Command Verbs as its current feature, with a clear statement that more exam-technique features are planned.
FR-41	MUST	The Command Verbs landing page must give access to four sub-features: Diagnostic, Verb-by-verb Guide, Practice, and Quick Reference.
FR-42	MUST	The Diagnostic must present 10–12 prompts that test the student's ability to identify command verbs and produce verb-appropriate responses.
FR-43	MUST	The Diagnostic must produce a per-verb strength report on completion.
FR-44	MUST	The Diagnostic must be repeatable, with different prompts on each attempt, and progress tracked across attempts.
FR-45	MUST	The Verb-by-verb Guide must include one page per verb (minimum: describe, explain, analyse, evaluate, justify, discuss, compare, assess) containing definition, AO-level expectations, worked example, common mistakes, and a 2–3 question mini-exercise.
FR-46	MUST	The Practice mode must present DSD-flavoured prompts and accept written or voice-input responses.
FR-47	MUST	Practice feedback must explicitly reference the command verb and at least one AO.
FR-48	MUST	The Quick Reference must present all verbs on a single page suitable for last-minute revision.
FR-49	SHOULD	Each verb page should display the student's current mastery level for that verb.

6.6 ExplainIT

ExplainIT is the distinctive teach-the-AI feature of StudyNest. The student selects a topic and engages in a probing conversation with an AI that pretends not to understand, asking follow-up questions until the student demonstrates genuine understanding.

ID	Priority	Requirement
FR-50	MUST	ExplainIT must be a top-level section accessible from the Student Dashboard.
FR-51	MUST	Students must be able to select any topic that has been authored with the necessary supporting content (misconceptions, probing questions) for ExplainIT to operate on.
FR-52	MUST	During a session, the AI character must adopt a consistent novice persona, expressing uncertainty and asking clarifying questions.
FR-53	MUST	The AI must hold misconceptions drawn from the curated misconception library, not generic novice misconceptions.
FR-54	MUST	The AI must pursue follow-up questions when the student's explanation is incomplete, vague, or surface-level — not accept the first plausible response.
FR-55	MUST	The session must end when one of: (a) the AI has been guided through all of its planned misconceptions and probing questions and reports understanding, (b) the student chooses to end the session, (c) a maximum turn-count is reached (configurable, default 20 turns).
FR-56	MUST	The full transcript of every ExplainIT session must be persisted, viewable by the student and (in anonymised form) by the admin.
FR-57	SHOULD	After session end, the student should receive a short reflective summary indicating which misconceptions were addressed and which (if any) remained.
FR-58	SHOULD	The student should be able to review their past ExplainIT transcripts.
FR-59	SHOULD	Voice input via Web Speech API must be available in ExplainIT sessions.
FR-60	SHOULD	The system should display a 'Sam's understanding' indicator (or similarly named) showing how close the AI character is to grasping the topic, updated as misconceptions are resolved.
FR-61	COULD	Students could optionally rate the helpfulness of an ExplainIT session at the end, contributing to feedback for prompt and content improvement.

6.7 Revision

ID	Priority	Requirement
FR-70	MUST	The Revision section must offer three activities: Mixed Quizzes, Past-Question-Style Practice, and the Spaced-Repetition Queue.
FR-71	MUST	Mixed Quizzes must draw questions across topics, with question selection biased toward topics where the student's mastery is lower.
FR-72	MUST	Past-Question-Style Practice must offer extended-response questions in the style of past papers, mapped to the same POs and command verbs, without verbatim reproduction of Pearson materials.
FR-73	MUST	The Spaced-Repetition Queue must surface items the student has previously attempted, prioritised by an algorithm combining previous score and time-since-last-review.

ID	Priority	Requirement
FR-74	SHOULD	Voice input via Web Speech API must be available for extended-response questions in Past-Question-Style Practice.

6.8 Admin Dashboard and student management

ID	Priority	Requirement
FR-80	MUST	The Admin Dashboard must display, at a glance: total student count, active student count (signed in within the last 7 days), recent activity highlights, and links to the main admin functions.
FR-81	MUST	Admins must be able to view a list of all cohorts and the students in each.
FR-82	MUST	Admins must be able to create a new cohort with a name and academic year.
FR-83	MUST	Admins must be able to create a new student account, specifying handle, initial password, cohort, and (optionally) real name. Real name visible only to admins, never to other students.
FR-84	MUST	Admins must be able to bulk-create student accounts from an uploaded CSV file.
FR-85	MUST	Admins must be able to view aggregate cohort progress: average mastery per topic, distribution of scores, students who have not signed in recently.
FR-86	MUST	Admins must be able to view an individual student's detail page (mastery, recent activity, sessions completed, last sign-in).
FR-87	MUST	Admins must be able to reset a student's password (forcing a change on the student's next sign-in).
FR-88	MUST	Admins must be able to deactivate a student account (preserves data, prevents sign-in).
FR-89	MUST	Admins must be able to delete a student account (with two-step confirmation, anonymising historical data rather than removing it from analytics).
FR-90	SHOULD	The admin view should highlight students who appear to be struggling (low mastery, infrequent use, recent decline).
FR-91	SHOULD	The admin view should highlight students who appear to be over-using AI feedback features (potential over-reliance).
FR-92	SHOULD	Admins should be able to export cohort progress data as CSV for offline analysis.
FR-93	COULD	Admins could send broadcast announcements visible on the Student Dashboard.
FR-94	LATER	Admins could later assign specific topics or activities as homework with due dates.

6.9 Cross-cutting requirements

ID	Priority	Requirement
FR-100	MUST	Mastery scores must update automatically based on attempt results, using a defined scoring algorithm.
FR-101	MUST	The spaced-repetition queue must be updated automatically when an attempt is recorded, recalculating next-review dates.
FR-102	MUST	All LLM calls must be made server-side; no API key may ever be exposed to the browser.
FR-103	MUST	The system must enforce per-user rate limits on LLM-backed features to manage cost and provider quota.
FR-104	MUST	All AI feedback must be clearly labelled as AI-generated draft feedback, not as a final mark.
FR-105	MUST	The system must function (with reduced features) when the LLM provider is unavailable — multiple-choice quizzes, content reading, and progress views must remain operational.

7. Non-functional requirements

Non-functional requirements describe how the system must behave — its performance characteristics, reliability, and quality attributes.

7.1 Performance

ID	Requirement
NFR-01	Page load time for any non-LLM page must be under 1.5 seconds on a typical college network.
NFR-02	Quiz feedback for auto-graded multiple-choice questions must appear within 200ms of submission.
NFR-03	AI-generated feedback for short-answer and extended-response questions must appear within 8 seconds, or display a clear progress indicator.
NFR-04	ExplainIT AI responses must appear within 5 seconds of student message submission, or display a typing indicator.
NFR-05	The system must support at least 30 concurrent active student users without performance degradation.

7.2 Reliability and availability

ID	Requirement
NFR-06	The system must be available 99% of the time during the UK school day (08:00–17:00) during term time.
NFR-07	Planned maintenance must occur outside the UK school day where possible.
NFR-08	When the LLM provider is unavailable, the system must degrade gracefully: clear messaging rather than error pages, non-LLM features remaining operational.

7.3 Scalability

ID	Requirement
NFR-09	The architecture must support scaling from a single college (100 students) to multiple colleges (up to 1,000 students) without fundamental redesign.
NFR-10	Content must scale to at least 50 topics across Core Paper 1 and Core Paper 2 without performance degradation.
NFR-11	Database queries must remain performant with at least 100,000 attempt records (roughly two academic years of cohort use).

7.4 Maintainability

ID	Requirement
NFR-12	Topic content must be authorable in plain Markdown by anyone comfortable with basic text editing and Git.
NFR-13	Quiz content must be authorable in YAML against a documented schema, validated automatically before deployment.
NFR-14	Adding a new topic must not require any code changes — only adding content files in the correct location.
NFR-15	Adding a new command verb must require only a single Markdown file plus an entry in the verb-prompts YAML.
NFR-16	All system prompts used by LLM features (including ExplainIT personas and probing strategies) must be version-controlled in the repository.

7.5 Cost

ID	Requirement
NFR-17	All external services used in the first release must be on free tiers at pilot scale (one cohort up to 30 students).
NFR-18	Total monthly running cost must not exceed £20 at pilot scale, including any optional paid services.
NFR-19	If LLM costs at scale exceed free-tier limits, the system must support a swap to a different LLM provider via configuration only — no code changes.

7.6 Browser and device support

ID	Requirement
NFR-20	The system must function correctly in current versions of Chrome, Firefox, Safari, and Edge.
NFR-21	The system must be fully usable on mobile phones, with all features accessible (touch targets ≥ 44px, no horizontal scrolling, readable typography at default zoom).
NFR-22	The system must be usable on tablets in both portrait and landscape orientations.
NFR-23	Voice input must function on browsers supporting the Web Speech API; where unavailable, text input must remain a fallback.

8. Content management strategy

Content in StudyNest is split between two locations based on type. The split is deliberate and is the central architectural decision underpinning maintainability.

8.1 What lives where

Content type	Location	Why
Topic prose (lessons)	Markdown files in the repository	Long-form text edits comfortably in any editor; version control matters; portable.
Quiz questions	YAML files in the repository	Repeating structure benefits from schema; spot-checking many questions easier in one file.
Command-verb guides	Markdown files in the repository	Same reasoning as topic prose.
Command-verb practice prompts	YAML files in the repository	Repeating structure.
Misconceptions library	YAML files in the repository	Structured data with topic links. Used by ExplainIT.
ExplainIT persona prompts and probing strategies	Plain text files under /prompts in the repository	Prompt engineering needs version control and review.
Specification (POs, AOs)	YAML in repo, mirrored to DB on startup	Source of truth in file; queryable mirror in DB.
Student data (attempts, mastery, sessions, transcripts)	Postgres database only	Per-user, per-attempt, queryable. Cannot live in files.
User accounts (handles, hashed passwords, roles)	Postgres database only	Authentication-critical data.
Cohorts and assignments	Postgres database only	Created and edited via Admin UI; not appropriate for files.

8.2 Authoring workflow

1. Author writes a topic in Markdown, in any text editor, with the prescribed frontmatter.
2. Author writes the accompanying quiz YAML file using the documented schema.
3. Validation script runs locally to confirm the files conform to schema.
4. Files are committed to Git and pushed to the main branch.
5. CI runs the validation suite; on success, deployment proceeds.
6. On application startup, the database mirror tables (topics, questions, command_verbs, misconceptions) are synchronised from the file content.

8.3 Markdown topic file format

Each topic file lives at content/topics/{paper}/{slug}.md with this structure:

```
---
title: Requirements Analysis
core_paper: 1
topic_number: 2
performance_outcomes: [PO1, PO3]
estimated_minutes: 25
explainit_enabled: true
last_updated: 2026-09-01
---

# Requirements Analysis

Lesson content in Markdown.

## Functional vs Non-functional requirements

A **functional requirement** describes...

## Worked example

Consider a school library system...
```

8.4 YAML quiz file format

Each quiz file lives at content/quizzes/{paper}/{slug}-quiz.yaml and conforms to a documented schema. See the original draft for a worked example; this remains unchanged.

8.5 The Admin UI does not author content

The Admin UI is for managing students, cohorts, and analytics — not for authoring topics, quizzes, command-verb content, misconceptions, or ExplainIT prompts. Those remain in version-controlled files. This separation keeps content authoring fast and transparent, while making student management database-backed and accessible from any device.

9. Data model and entity-relationship design

Persisted entities, their attributes, and relationships. Some entities are mirrors of file-based content (topics, questions, command_verbs, misconceptions); others are exclusively database-resident (users, attempts, mastery, sessions, cohorts, transcripts).

9.1 Entities and attributes

9.1.1 cohorts

Attribute	Description
id	UUID, primary key
name	Text. e.g. 'Strode DSD Year 2 2026/27'
academic_year	Text. e.g. '2026/27'
created_by	UUID, foreign key to users.id
created_at	Timestamp

9.1.2 users

Attribute	Description
id	UUID, primary key
handle	Text, unique. Pseudonymous identifier shown in the system.
real_name	Text, nullable. Visible only to admins.
password_hash	Text. Bcrypt or Argon2 hashed.
role	Enum: 'student', 'admin'
cohort_id	UUID, foreign key to cohorts.id, nullable (admins may have no cohort)
must_change_password	Boolean. True for newly created accounts and after password resets.
is_active	Boolean. False for deactivated accounts.
last_signin_at	Timestamp, nullable
created_at	Timestamp
updated_at	Timestamp

9.1.3 specifications

Attribute	Description
id	UUID, primary key
name	Text. e.g. 'DSD Core Paper 1'

Attribute	Description
version	Text. The specification version this maps to.
loaded_at	Timestamp. When this was last synced from YAML source.

9.1.4 performance_outcomes

Attribute	Description
id	UUID, primary key
specification_id	UUID, foreign key to specifications.id
code	Text. e.g. 'PO1', 'PO3'
title	Text
description	Text

9.1.5 assessment_objectives

Attribute	Description
id	UUID, primary key
code	Text. 'AO1', 'AO2', 'AO3'
description	Text

9.1.6 topics

Mirror of topic Markdown files. Synced from /content/topics on startup.

Attribute	Description
id	UUID, primary key
slug	Text, unique. Filename-derived.
title	Text
core_paper	Integer. 1 or 2.
topic_number	Integer. Order within paper.
estimated_minutes	Integer
overview	Text. First-paragraph summary.
explainit_enabled	Boolean. Whether ExplainIT supports this topic (requires a populated misconception library).
content_path	Text. Path to source Markdown file.
last_updated	Date. From frontmatter.
synced_at	Timestamp.

9.1.7 topic_po_links

Attribute	Description
topic_id	UUID, foreign key to topics.id
performance_outcome_id	UUID, foreign key to performance_outcomes.id

9.1.8 command_verbs

Attribute	Description
id	UUID, primary key
verb	Text, unique. Lowercase.
tier	Enum: 'recall', 'apply', 'analyse'
definition	Text
expectations_by_ao	JSON. Keyed by AO code.
content_path	Text. Path to source Markdown file.

9.1.9 questions

Mirror of quiz YAML files.

Attribute	Description
id	UUID, primary key
external_id	Text, unique. The 'id' field from YAML.
topic_id	UUID, foreign key to topics.id, nullable
command_verb_id	UUID, foreign key to command_verbs.id, nullable
type	Enum: 'multiple_choice', 'short_answer', 'extended_response'
prompt	Text
options	JSON, nullable
correct_index	Integer, nullable
model_answer	Text, nullable
mark_scheme	JSON, nullable
explanation	Text, nullable
performance_outcome_id	UUID, foreign key to performance_outcomes.id
assessment_objective_id	UUID, foreign key to assessment_objectives.id
synced_at	Timestamp

9.1.10 misconceptions

Mirror of misconceptions YAML files. Used by ExplainIT to populate the AI character's mental model.

Attribute	Description
id	UUID, primary key
external_id	Text, unique
topic_id	UUID, foreign key to topics.id
description	Text. The mistaken belief.
typical_response_pattern	Text. How a student holding this misconception typically writes.
probing_questions	JSON. Array of follow-up questions the AI asks to surface this misconception.
correct_understanding	Text. The accurate framing the student needs to convey for the misconception to be 'resolved'.

9.1.11 sessions

Attribute	Description
id	UUID, primary key
user_id	UUID, foreign key to users.id
type	Enum: 'topic_quiz', 'mixed_quiz', 'past_question_practice', 'spaced_repetition', 'explainit', 'command_verbs_diagnostic', 'command_verbs_practice'
context_id	UUID, nullable. e.g. topic_id for a topic_quiz or explainit session.
started_at	Timestamp
ended_at	Timestamp, nullable
summary	JSON. Per-type summary data.

9.1.12 attempts

Attribute	Description
id	UUID, primary key
session_id	UUID, foreign key to sessions.id
user_id	UUID, foreign key to users.id (denormalised for query speed)
question_id	UUID, foreign key to questions.id
response_text	Text
response_voice_used	Boolean
score	Numeric. Normalised 0–1.
max_score	Integer

Attribute	Description
feedback	JSON. Structured feedback (verb, AO, points hit/missed, suggestions).
llm_provider	Text, nullable
llm_tokens_used	Integer, nullable
attempted_at	Timestamp

9.1.13 mastery

Attribute	Description
id	UUID, primary key
user_id	UUID, foreign key to users.id
topic_id	UUID, foreign key to topics.id, nullable
command_verb_id	UUID, foreign key to command_verbs.id, nullable
score	Numeric. Current mastery estimate, 0–1.
attempts_count	Integer
last_practised_at	Timestamp
next_review_at	Timestamp

A row exists for each (user, topic) pair OR for each (user, command_verb) pair. Exclusivity enforced at write-time: exactly one of topic_id and command_verb_id is non-null.

9.1.14 explainit_transcripts

One row per ExplainIT session. Stores the full conversation and the state of the AI character's understanding at session end.

Attribute	Description
id	UUID, primary key
session_id	UUID, foreign key to sessions.id
user_id	UUID, foreign key to users.id
topic_id	UUID, foreign key to topics.id
misconceptions_planned	JSON. Array of misconception ids the AI was set up to hold at session start.
misconceptions_resolved	JSON. Array of misconception ids the student successfully addressed during the session.
transcript	JSON. Array of {speaker, text, timestamp} entries.
completed	Boolean. True if all planned misconceptions were resolved.
end_reason	Enum: 'completed', 'student_ended', 'turn_limit'

Attribute	Description
student_rating	Integer, nullable. 1–5 if the student rated the session.
created_at	Timestamp

9.2 Entity-relationship summary

```
cohorts —< users
users —< sessions —< attempts
users —< mastery
users —< explainit_transcripts

specifications —< performance_outcomes
specifications —< topics
topics >—< performance_outcomes (via topic_po_links)
topics —< questions
topics —< misconceptions
topics —< explainit_transcripts

command_verbs —< questions
performance_outcomes —< questions
assessment_objectives —< questions

sessions —< explainit_transcripts (1:1 for ExplainIT sessions)
```

9.3 Indexing

- users.handle — unique index, sign-in lookup.
- attempts.user_id, attempts.attempted_at — composite, per-user activity queries.
- attempts.question_id — question-level analytics.
- mastery.user_id, mastery.next_review_at — composite, spaced-repetition queue.
- sessions.user_id, sessions.started_at — composite, recent activity queries.
- explainit_transcripts.user_id, explainit_transcripts.created_at — for past-sessions list.
- topics.slug — unique.
- questions.external_id — unique.
- misconceptions.external_id — unique.

10. System architecture

10.1 High-level architecture

StudyNest is a single Flask application serving server-rendered pages, with progressive enhancement via HTMX for dynamic interactions and a small amount of vanilla JavaScript for browser-only features (voice input, ExplainIT live conversation updates).

```
Browser
    │  HTTPS
    ▼
Flask application (Python)
    ├── Jinja2 templates (HTML rendering)
    ├── HTMX (dynamic partial updates)
    ├── Tailwind CSS (styling)
    ├── Routes layer
    │       ├── public (landing, sign-in)
    │       ├── student (dashboard, core_paper, exam_technique, explainit, revision)
    │       └── admin (admin dashboard, student mgmt, analytics, settings)
    ├── Services layer
    │       ├── content_loader
    │       ├── llm_client
    │       ├── scoring
    │       ├── spaced_repetition
    │       └── explainit_engine (persona, probing logic, state)
    └── Models layer (SQLAlchemy)
    ▼
Postgres database (Supabase)

External services:
    ├── Supabase (Postgres)
    ├── Google Gemini API (primary LLM)
    ├── Groq API (fallback LLM)
    └── Web Speech API (browser-side, voice input)
```

10.2 Stack summary

Layer	Choice	Notes
Backend framework	Flask 3.x	Server-rendered, lightweight, Python-native.
Templating	Jinja2	Built into Flask.
Dynamic interactions	HTMX (CDN)	No build step.
Styling	Tailwind CSS	Via CDN initially.
Database	Postgres on Supabase free tier	500MB / 50k MAU free.
ORM	SQLAlchemy 2.x	Standard Python ORM.
Auth	Flask-Login + bcrypt	Custom user table.
LLM (primary)	Google Gemini Flash	Generous current free tier.
LLM (fallback)	Groq	Open-weight models, fast.

Layer	Choice	Notes
Voice input	Web Speech API	Browser-native, free.
Hosting	Render or Fly.io free tier	Both support Flask.
Repo & CI	GitHub + GitHub Actions	Free for public repos.
Production server	Gunicorn	Standard WSGI.

10.3 Project folder structure

```
studynest/
├── app/
│   ├── __init__.py
│   ├── routes/
│   │   ├── public.py
│   │   ├── auth.py
│   │   ├── dashboard.py
│   │   ├── core_paper.py
│   │   ├── revision.py
│   │   ├── exam_technique.py
│   │   ├── explainit.py
│   │   └── admin.py
│   ├── models/
│   ├── services/
│   │   ├── content_loader.py
│   │   ├── llm_client.py
│   │   ├── scoring.py
│   │   ├── spaced_repetition.py
│   │   └── explainit_engine.py
│   ├── templates/
│   │   ├── base.html
│   │   ├── public/landing.html
│   │   ├── student/dashboard.html
│   │   ├── admin/dashboard.html
│   │   └── partials/
│   └── static/
├── content/
│   ├── topics/
│   ├── quizzes/
│   ├── command_verbs/
│   ├── command_verb_prompts.yaml
│   ├── misconceptions.yaml
│   └── specification.yaml
├── prompts/
│   ├── extended_response_feedback.txt
│   ├── explainit_persona.txt
│   ├── explainit_probing_strategy.txt
│   └── short_answer_grading.txt
├── tests/
├── scripts/
├── migrations/
├── requirements.txt
├── .env.example
├── config.py
├── run.py
└── README.md
```

11. External integrations

11.1 LLM provider

- Primary: Google Gemini Flash (free tier).
- Fallback: Groq (free tier, open-weight models).
- Provider switch by configuration only.
- System prompts versioned in /prompts.
- All LLM calls log: provider, model, tokens, latency, request id.

11.2 ExplainIT-specific LLM use

ExplainIT places higher demands on the LLM than other features:

- Persona consistency across many turns within a session.
- Stateful tracking of which misconceptions remain unresolved.
- Pursuit of follow-up questions rather than acceptance of surface-level answers.

These must be implemented through a combination of careful system prompts, structured turn-state passed back to the model on each call, and explicit checks in the explainit_engine service rather than relying on the LLM alone.

11.3 Database

- Postgres on Supabase free tier.
- Migrations via Alembic.
- Daily automated backup retained 7 days (Supabase free-tier feature).

11.4 Web Speech API

- Browser-side only; no server speech processing.
- Used in: Command Verbs Practice, ExplainIT, Past-Question-Style Practice extended responses.
- Text input fallback where unavailable.
- No audio data is transmitted to or stored on the server.

12. Security, privacy and compliance

12.1 Authentication

- Passwords hashed with bcrypt or Argon2.
- Sessions: secure, HttpOnly, SameSite=Strict cookies.
- CSRF protection on state-changing requests.
- Sign-in rate-limited (5 attempts per minute per IP).
- Forced password change on first sign-in and after admin reset.

12.2 Authorisation

- All routes authenticated except Landing and sign-in.
- Role-based access enforced server-side.
- Students access only their own data.
- Admin routes accessible only to role='admin'.
- Cross-account URL access returns 403.

12.3 Privacy

- Pseudonymous handles are the primary identifier in cohort views.
- Real names optional, visible only to admins, never sent to LLM providers.
- Student responses sent to LLM providers must not include identifying information beyond the response itself.
- LLM provider data retention and training-use policies must be reviewed and documented before launch.
- Privacy notice presented at first sign-in, requiring acknowledgement.

12.4 Data protection (UK GDPR / DPA 2018)

- Lawful basis: to be confirmed with college DPO before launch.
- Data minimisation: only data needed to operate StudyNest is collected.
- Right to erasure: an account can be deactivated and historical data anonymised on request.
- Data export: a student can request export of their own data.
- Cross-border data transfer: LLM providers may process outside the UK; documented in privacy notice and approved by DPO.

12.5 Safeguarding

- Some users will be under 18.
- No student-to-student content sharing in StudyNest.

- AI-generated content must pass safety filters appropriate to the audience.
- Concerning patterns (e.g. self-harm references in free-text responses) must be flagged for human review. Implementation specifics to be agreed with safeguarding lead.

13. Accessibility requirements

StudyNest must be accessible to students with a range of needs. Many DSD students have specific learning differences, EAL backgrounds, or sensory differences. Accessibility is built in, not bolted on.

13.1 Standards

WCAG 2.2 AA baseline.

13.2 Specific requirements

- All interactive elements keyboard-accessible with visible focus.
- Colour contrast meets AA ratios; no information conveyed by colour alone.
- Non-decorative images have meaningful alt text.
- Headings form a logical, sequential outline.
- Forms have correctly associated labels.
- Voice input available for any extended-response field.
- Default typography uses or supports a dyslexia-friendly font (e.g. Atkinson Hyperlegible).
- Body text minimum 16px; line-length maximum approximately 70 characters.
- High-contrast theme available.
- No auto-submit on inactivity; all timers pausable.
- Animations respect prefers-reduced-motion.

13.3 Testing

- Automated accessibility scanning (axe-core) in CI.
- Manual keyboard navigation testing of critical flows before each release.
- Screen reader smoke-testing of Dashboard, a topic page, a quiz, and an ExplainIT session before each release.

14. Key performance indicators

KPIs organised by goal. Each has a measurement method and a target where reasonably set.

14.1 Goal 1 — Student outcomes

ID	KPI	Measurement	Target
KPI-01	Cohort grade distribution in Core Paper 1 and 2	Compare exam results of cohorts using StudyNest against historical baselines.	After 1 year: positive shift in grade distribution.
KPI-02	Mastery score growth	Mean topic-mastery per active student, weekly across the year.	Mean mastery > 0.7 by exam-revision phase.
KPI-03	Command-verb diagnostic improvement	First vs latest Diagnostic score per student.	Mean improvement ≥ 25% by mid-year.
KPI-04	Voluntary use	Proportion of cohort signing in weekly without homework being set.	≥ 60% by mid-year.
KPI-05	Topic coverage at exam time	Proportion of spec topics where student has completed quiz at least once, in the month before exams.	Cohort median > 80%.
KPI-06	ExplainIT engagement	Proportion of cohort completing at least 5 ExplainIT sessions across the year.	≥ 50%.

14.2 Goal 2 — Software engineering portfolio value

ID	KPI	Measurement	Target
KPI-07	Codebase quality	Test coverage, lint compliance, documentation completeness.	Coverage ≥ 60% at first release; lint clean; README and CONTRIBUTING present.
KPI-08	Public visibility	Repo public on GitHub, README readable, deployed live URL accessible.	All present at first release.
KPI-09	Student contributions	Number of students with at least one merged PR.	Aspirational: ≥ 3 by end of academic year.
KPI-10	Industry-standard workflow	Non-trivial changes merged via PR with at least one review.	Process observed throughout development.

14.3 Goal 3 — Recognition and reach

ID	KPI	Measurement	Target
KPI-11	Adoption beyond Strode	Number of FE colleges using StudyNest with at least one cohort.	Year 1: ≥ 1 other institution.
KPI-12	Practitioner visibility	Mentions in FE digital networks, talks, blog posts, citations.	Year 1: ≥ 1 talk or post.
KPI-13	Articulable philosophy	External practitioners can describe StudyNest in one sentence.	Validated qualitatively.

14.4 Operational KPIs

ID	KPI	Measurement	Target
KPI-14	Uptime	% availability during UK school day, term time.	≥ 99%.
KPI-15	LLM cost per active student per month	Total LLM spend / unique active students.	≤ £0.50 / student / month.
KPI-16	Page load p95	95th percentile page load on typical college network.	≤ 2.0s.
KPI-17	AI feedback quality	% of sampled AI feedback rated 'pedagogically sound' on monthly review.	≥ 90%.
KPI-18	ExplainIT persona consistency	% of sampled sessions where AI character maintains consistent novice persona.	≥ 90%.

15. Acceptance criteria for the MVP

StudyNest is considered to have shipped its first release when all the following criteria are met.

15.1 Functional acceptance

7. An admin can sign in, create a cohort, and create at least 30 student accounts (individually or via bulk import).
8. A student can sign in with credentials provided by the admin, change their password on first sign-in, and reach the Student Dashboard.
9. The Student Dashboard shows a personalised greeting, a recommended next session, a mastery overview, and weak spots — all with real data.
10. At least 5 Core Paper 1 topics fully implemented, each with concept content, topic quiz, and at least one extended-response question with command-verb-aware feedback.
11. Exam Technique → Command Verbs is fully built: diagnostic, all 8 verb guides, practice mode with feedback, quick reference.
12. ExplainIT operational on at least 3 topics, with curated misconception libraries for each. The AI character probes consistently and the session terminates correctly under all three end conditions.
13. Revision section operational with mixed quizzes and spaced-repetition queue. Past-question practice functional, even if content depth limited.
14. Voice input works in at least one feature on Chromium browsers.
15. Admin Dashboard: cohort overview, student detail, password reset, deactivation, and delete are all functional.

15.2 Quality acceptance

16. All MUST functional requirements implemented.
17. Test coverage ≥ 60% on application logic.
18. Lighthouse accessibility score ≥ 90 on all main pages.
19. Manual keyboard-only navigation completes a full session (sign-in, take a quiz, do an ExplainIT session, sign-out).
20. All AI feedback prompts spot-checked on at least 20 sample responses each.
21. All ExplainIT misconception libraries reviewed by Mario for pedagogical fidelity.
22. Privacy notice and DPO sign-off complete.
23. README documents how to run locally, deploy, and add new content.

15.3 Operational acceptance

24. Deployed live at a stable URL.

25. CI/CD running on every push to main.
26. Database backups configured and one restore verified.
27. Cost projection validated against actual measurements over a one-week test.
28. Monitoring or simple uptime check in place.

16. Risks and mitigations

Risk	Likelihood / Impact	Mitigation
Pearson intellectual property concerns	Medium / High	Direct conversation with Pearson before launch. Build practice content in the style of, not verbatim from, official materials.
AI feedback plausible but pedagogically wrong	High / High	Mario reviews feedback patterns before they ship. Prompts version-controlled. AI feedback always labelled as draft.
ExplainIT AI character breaks persona or accepts surface-level answers	High / High	Persona enforced via system prompt with explicit constraints. Misconception state tracked in code, not in LLM. KPI-18 monitors persona consistency on a sampled basis.
LLM provider free-tier limits hit at scale	Medium / Medium	Provider abstraction allows swap by config. Aggressive caching. Per-user rate limits. Local model fallback option.
Content authoring stalls under teaching workload	High / High	Authoring done in lockstep with lesson preparation. Markdown-based authoring keeps friction low.
Over-reliance by students; undermined independent thinking	Medium / Medium	Admin view surfaces over-reliance patterns. Tool design encourages attempt-before-feedback.
Safeguarding concerns from under-18 students	Low / High	Pseudonymous accounts. Privacy notice. DPO consultation. Concerning-content alerts.
College policy on AI tools	Medium / Medium	Conversation with Strode senior leader before student deployment.
Spec changes (DSD specification revision)	Low / Medium	Specification mapping in YAML. Versioning supported in data model.
Single-author bottleneck	Medium / Medium	Comprehensive documentation. Code commented. Authoring documented. Aspirational: at least one other contributor by year 2.

17. Out of scope for the first release

17.1 Deferred features

- ESP module — full content and project-companion functionality.
- OS module — concept tutor and OS-task practice.
- Additional Exam Technique features beyond Command Verbs (time management, structuring extended responses, reading the question, self-marking).
- Homework assignment from the admin view.
- Broadcast announcements from the admin view.
- Student-contributed content via the UI.
- A separate teacher role between student and admin.
- Multi-college / multi-tenant support.
- Single sign-on with college identity provider.
- Native mobile app.
- Offline mode.
- Learning analytics export to college systems.
- Parent / guardian views.

17.2 Deliberately not built

- Public leaderboards or peer comparison.
- Points or XP detached from learning outcomes.
- Punishing daily streaks.
- Any feature that lets one student see another student's content.
- A 'give me the answer' mode bypassing attempt-before-feedback.
- Admin-side study mode (admins use a separate student account if they want to use the study features).

18. Open questions and decisions to be made

18.1 Naming and identity

- StudyNest logo and visual identity beyond the basic palette.
- Domain name.
- ExplainIT character name (working: 'Sam').

18.2 Content and IP

- Pearson position on use of mark scheme principles. Confirm before mark-scheme-calibrated feedback ships.
- Whether to seek formal Pearson endorsement.

18.3 Policy and compliance

- Strode College policy on AI tooling for student use.
- DPO sign-off on data flows, particularly LLM data transfer.
- Safeguarding lead sign-off on flagging mechanisms.

18.4 Open source

- Open-source licence — MIT or Apache 2.0 likely; confirm before public repo.
- Contribution policy.

18.5 Technical decisions deferred to build time

- Tailwind via CDN or build pipeline.
- Specific LLM model versions.
- Hosting choice between Render and Fly.io.
- ExplainIT turn-state representation passed to the LLM.

19. Document control

19.1 Versioning

This document is versioned alongside the codebase. Substantive changes increment the version and are summarised in the change log.

19.2 Change log

Version	Date	Summary of changes
1.0	May 2026	Initial requirements specification.
2.0	May 2026	Product named StudyNest. Foundations renamed to Exam Technique. Teach the AI feature replaced by ExplainIT, promoted to top-level. Landing page and Dashboard separated; Student Dashboard and Admin Dashboard distinguished. User roles simplified from three (student/teacher/admin) to two (student/admin). Updated FRs, NFRs, data model, KPIs, and acceptance criteria throughout to reflect the above.
2.1	May 2026	Corrected DSD: Digital Software Development (not Digital Support and Development). Corrected awarding body: Pearson (not NCFE). Confirmed two-role model: admin and teacher are the same person at Strode; admin is the operative term throughout.

19.3 Review cadence

Reviewed at the end of each build phase. Items that proved unworkable, requirements that were missed, and assumptions that were invalidated should be reflected in revisions.